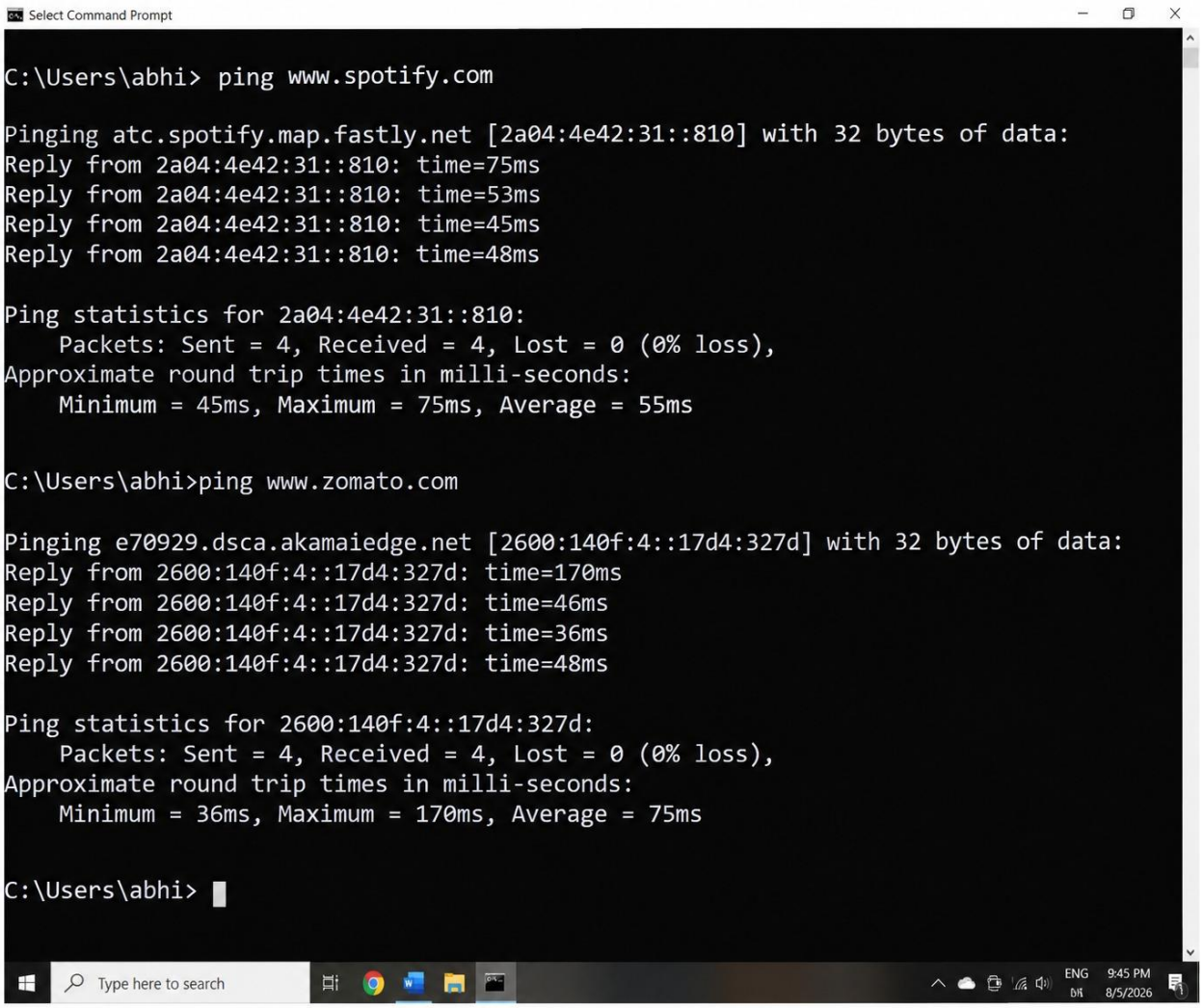
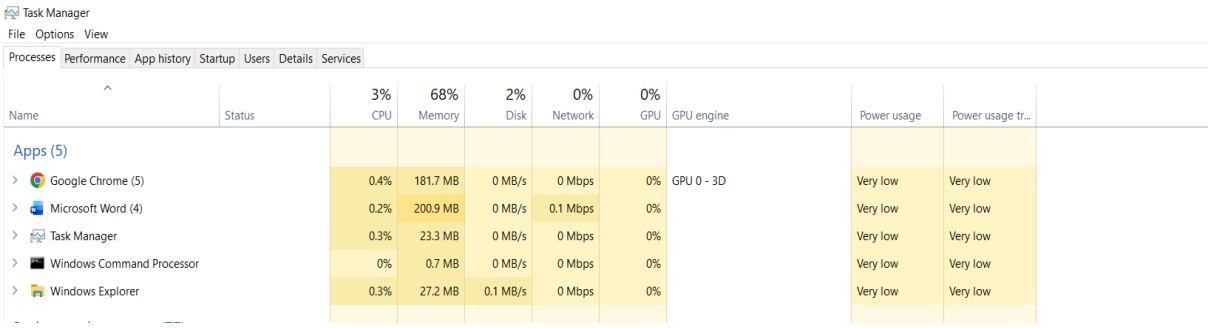
1.Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Protocol	Purpose
UDP (User Datagram Protocol)	Sends data quickly without checking if every packet arrives.
TLS (Transport Layer Security)	Encrypts data to make Internet communication secure (HTTPS).
DNS (Domain Name System)	Converts website names (like www.google.com) into IP addresses.

2.Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.



3.Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.



Application	Process Name	Memory Usage
Google Chrome	Google Chrome (5 processes)	181.7 MB
Microsoft Word	Microsoft Word (4 processes)	200.9 MB
Task Manager	Task Manager	23.3 MB
Windows Command Processor	Windows Command Processor	0.7 MB
Windows Explorer	Windows Explorer	27.2 MB

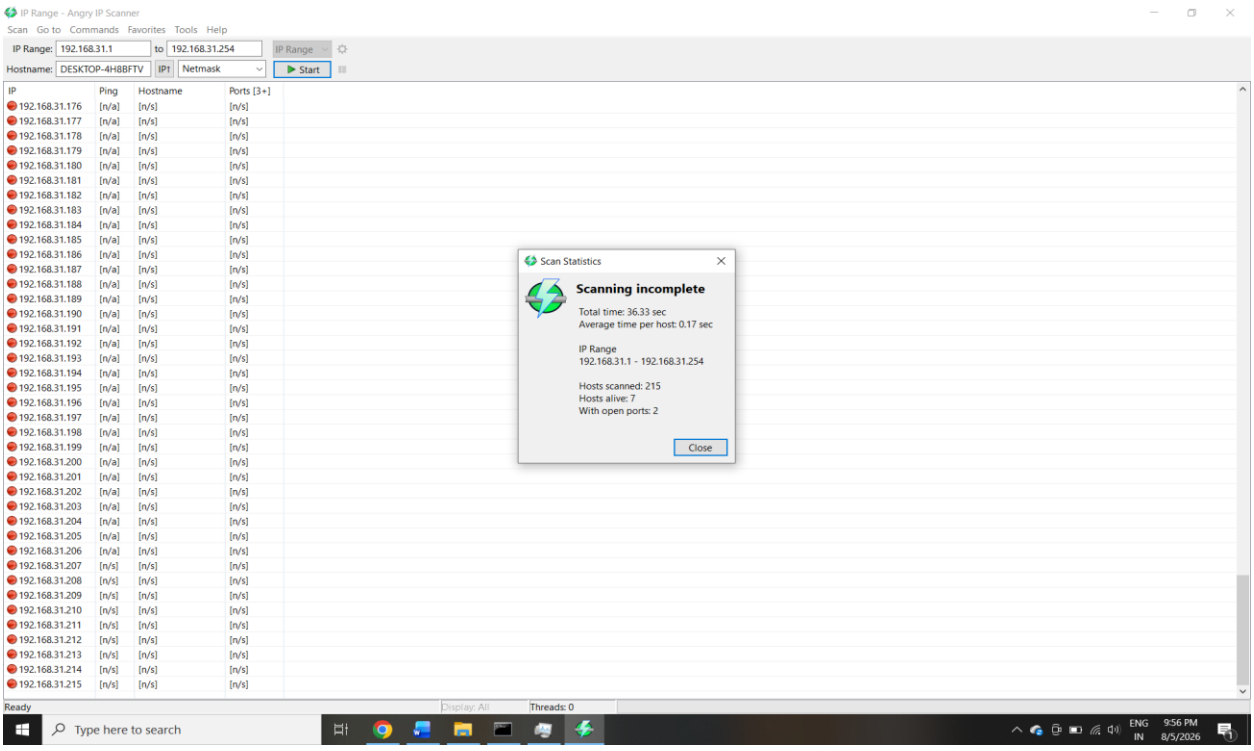
4.Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.Hint: Make sure your firewall allows scanning, and only scan your own home/college network.

Local WiFi Network Scan (Angry IP Scanner)

Tool Used: Angry IP Scanner

IP Range Scanned: 192.168.31.1 – 192.168.31.254

IP Address	Device Name / Hostname
192.168.31.1	jiofiber.local.html (JioFiber Router)
192.168.31.79	DESKTOP-4H8BFTV (My Windows Laptop)
192.168.31.103	SetTopBox-9929… (Jio Set-Top Box)



5.Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Hardware Tool	When I would use it
Cable Tester	Used to check whether an Ethernet cable is wired correctly and has no breaks or faults.
RJ45 Crimping Tool	Used to attach an RJ45 connector to a network cable when making or repairing Ethernet cables.
Digital Multimeter	Used to measure voltage, resistance, and continuity when troubleshooting power or network hardware issues.